

Recall from Section 4.5.3 that an autonomous system (AS) is a collection of routers under the same administrative and technical control, and that all run the same routing protocol among themselves. Each AS, in turn, typically contains multiple subnets (where we use the term subnet in the precise, addressing sense in Section 4.4.2).

4.6.1 Intra-AS Routing in the Internet: RIP

An intra-AS routing protocol is used to determine how routing is performed within an autonomous system (AS). Intra-AS routing protocols are also known as **interior gateway protocols**. Historically, two routing protocols have been used extensively for routing within an autonomous system in the Internet: the **Routing Information Protocol (RIP)** and **Open Shortest Path First (OSPF)**. A routing protocol closely related to OSPF is the **IS-IS** protocol [RFC 1142, Perlman 1999]. We first discuss RIP and then consider OSPF.

RIP was one of the earliest intra-AS Internet routing protocols and is still in widespread use today. It traces its origins and its name to the Xerox Network Systems (XNS) architecture. The widespread deployment of RIP was due in great part to its inclusion in 1982 in the Berkeley Software Distribution (BSD) version of UNIX supporting TCP/IP. RIP version 1 is defined in [RFC 1058], with a backward-compatible version 2 defined in [RFC 2453].

RIP is a distance-vector protocol that operates in a manner very close to the idealized DV protocol we examined in Section 4.5.2. The version of RIP specified in RFC 1058 uses hop count as a cost metric; that is, each link has a cost of 1. In the DV algorithm in Section 4.5.2, for simplicity, costs were defined between pairs of routers. In RIP (and also in OSPF), costs are actually from source router to a destination subnet. RIP uses the term *hop*, which is the number of subnets traversed along the shortest path from source router to destination subnet, including the destination subnet. Figure 4.34 illustrates an AS with six leaf subnets. The table in the figure indicates the number of hops from the source A to each of the leaf subnets.

The maximum cost of a path is limited to 15, thus limiting the use of RIP to autonomous systems that are fewer than 15 hops in diameter. Recall that in DV protocols, neighboring routers exchange distance vectors with each other. The distance vector for any one router is the current estimate of the shortest path distances from that router to the subnets in the AS. In RIP, routing updates are exchanged between neighbors approximately every 30 seconds using a **RIP response message**. The response message sent by a router or host contains a list of up to 25 destination subnets within the AS, as well as the sender's distance to each of those subnets. Response messages are also known as **RIP advertisements**.

Let's take a look at a simple example of how RIP advertisements work. Consider the portion of an AS shown in Figure 4.35. In this figure, lines connecting the routers denote subnets. Only selected routers (*A*, *B*, *C*, and *D*) and subnets (*w*, *x*, *y*,

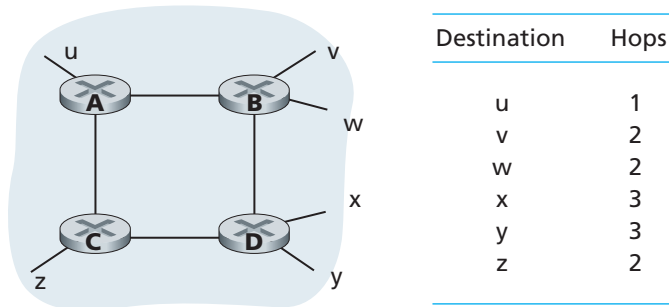


Figure 4.34 ♦ Number of hops from source router A to various subnets

and z) are labeled. Dotted lines indicate that the AS continues on; thus this autonomous system has many more routers and links than are shown.

Each router maintains a RIP table known as a **routing table**. A router's routing table includes both the router's distance vector and the router's forwarding table. Figure 4.36 shows the routing table for router D . Note that the routing table has three columns. The first column is for the destination subnet, the second column indicates the identity of the next router along the shortest path to the destination subnet, and the third column indicates the number of hops (that is, the number of subnets that have to be traversed, including the destination subnet) to get to the destination subnet along the shortest path. For this example, the table indicates that to send a datagram from router D to destination subnet w , the datagram should first be forwarded to neighboring router A ; the table also indicates that destination subnet w is two hops away along the shortest path. Similarly, the table indicates that subnet z is seven hops away via router B . In principle, a routing table will have one row for each subnet in the AS, although RIP version 2 allows subnet entries to be aggregated using route aggregation techniques similar to those we examined in

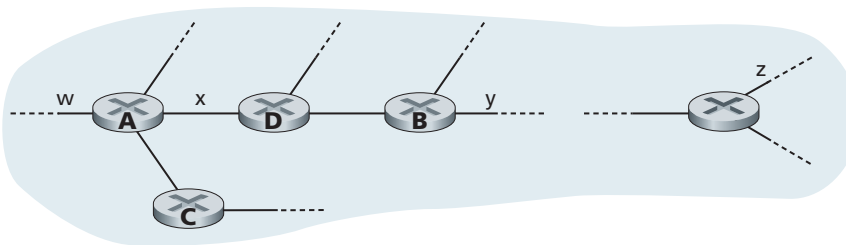


Figure 4.35 ♦ A portion of an autonomous system

Destination Subnet	Next Router	Number of Hops to Destination
w	A	2
y	B	2
z	B	7
x	—	1
. . . .		

Figure 4.36 ♦ Routing table in router *D* before receiving advertisement from router *A*

Section 4.4. The table in Figure 4.36, and the subsequent tables to come, are only partially complete.

Now suppose that 30 seconds later, router *D* receives from router *A* the advertisement shown in Figure 4.37. Note that this advertisement is nothing other than the routing table information from router *A*! This information indicates, in particular, that subnet *z* is only four hops away from router *A*. Router *D*, upon receiving this advertisement, merges the advertisement (Figure 4.37) with the old routing table (Figure 4.36). In particular, router *D* learns that there is now a path through router *A* to subnet *z* that is shorter than the path through router *B*. Thus, router *D* updates its routing table to account for the shorter shortest path, as shown in Figure 4.38. How is it, you might ask, that the shortest path to subnet *z* has become shorter? Possibly, the decentralized distance-vector algorithm is still in the process of converging (see Section 4.5.2), or perhaps new links and/or routers were added to the AS, thus changing the shortest paths in the AS.

Let’s next consider a few of the implementation aspects of RIP. Recall that RIP routers exchange advertisements approximately every 30 seconds. If a router does not hear from its neighbor at least once every 180 seconds, that neighbor is considered to be no longer reachable; that is, either the neighbor has died or the

Destination Subnet	Next Router	Number of Hops to Destination
z	C	4
w	—	1
x	—	1
. . . .		

Figure 4.37 ♦ Advertisement from router *A*

Destination Subnet	Next Router	Number of Hops to Destination
w	A	2
y	B	2
z	A	5
....		

Figure 4.38 ♦ Routing table in router *D* after receiving advertisement from router *A*

connecting link has gone down. When this happens, RIP modifies the local routing table and then propagates this information by sending advertisements to its neighboring routers (the ones that are still reachable). A router can also request information about its neighbor's cost to a given destination using RIP's request message. Routers send RIP request and response messages to each other over UDP using port number 520. The UDP segment is carried between routers in a standard IP datagram. The fact that RIP uses a transport-layer protocol (UDP) on top of a network-layer protocol (IP) to implement network-layer functionality (a routing algorithm) may seem rather convoluted (it is!). Looking a little deeper at how RIP is implemented will clear this up.

Figure 4.39 sketches how RIP is typically implemented in a UNIX system, for example, a UNIX workstation serving as a router. A process called *routed* (pronounced “route dee”) executes RIP, that is, maintains routing information and exchanges messages with *routed* processes running in neighboring routers. Because RIP is implemented as an application-layer process (albeit a very special one that is able to

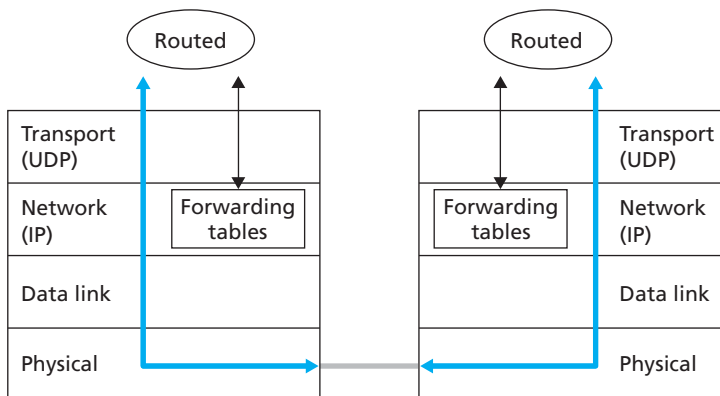


Figure 4.39 ♦ Implementation of RIP as the *routed* daemon

manipulate the routing tables within the UNIX kernel), it can send and receive messages over a standard socket and use a standard transport protocol. As shown, RIP is implemented as an application-layer protocol (see Chapter 2) running over UDP. If you're interested in looking at an implementation of RIP (or the OSPF and BGP protocols that we will study shortly), see [Quagga 2012].

4.6.2 Intra-AS Routing in the Internet: OSPF

Like RIP, OSPF routing is widely used for intra-AS routing in the Internet. OSPF and its closely related cousin, IS-IS, are typically deployed in upper-tier ISPs whereas RIP is deployed in lower-tier ISPs and enterprise networks. The Open in OSPF indicates that the routing protocol specification is publicly available (for example, as opposed to Cisco's EIGRP protocol). The most recent version of OSPF, version 2, is defined in RFC 2328, a public document.

OSPF was conceived as the successor to RIP and as such has a number of advanced features. At its heart, however, OSPF is a link-state protocol that uses flooding of link-state information and a Dijkstra least-cost path algorithm. With OSPF, a router constructs a complete topological map (that is, a graph) of the entire autonomous system. The router then locally runs Dijkstra's shortest-path algorithm to determine a shortest-path tree to all *subnets*, with itself as the root node. Individual link costs are configured by the network administrator (see Principles and Practice: Setting OSPF Weights). The administrator might choose to set all link costs to 1, thus achieving minimum-hop routing, or might choose to set the link weights to be inversely proportional to link capacity in order to discourage traffic from using low-bandwidth links. OSPF does not mandate a policy for how link weights are set (that is the job of the network administrator), but instead provides the mechanisms (protocol) for determining least-cost path routing for the given set of link weights.

With OSPF, a router broadcasts routing information to *all* other routers in the autonomous system, not just to its neighboring routers. A router broadcasts link-state information whenever there is a change in a link's state (for example, a change in cost or a change in up/down status). It also broadcasts a link's state periodically (at least once every 30 minutes), even if the link's state has not changed. RFC 2328 notes that "this periodic updating of link state advertisements adds robustness to the link state algorithm." OSPF advertisements are contained in OSPF messages that are carried directly by IP, with an upper-layer protocol of 89 for OSPF. Thus, the OSPF protocol must itself implement functionality such as reliable message transfer and link-state broadcast. The OSPF protocol also checks that links are operational (via a HELLO message that is sent to an attached neighbor) and allows an OSPF router to obtain a neighboring router's database of network-wide link state.

Some of the advances embodied in OSPF include the following:

- *Security.* Exchanges between OSPF routers (for example, link-state updates) can be authenticated. With authentication, only trusted routers can participate